

Secure Rate-Distortion-Perception Trade-off Over Channels: A Randomized Distributed Function Computation (RDFC) Application

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Abstract—Secure rate-distortion-perception (RDP) trade-offs arise in critical applications, such as semantic compression and privacy-preserving generative coding, where preserving perceptual quality while minimizing distortion is vital. This paper studies a framework for secure RDP over noiseless and noisy broadcast channels under strong secrecy constraints. We first characterize the exact secure RDP region for noiseless transmission channels. We then develop an inner bound on the secure RDP region for a memoryless broadcast channel with correlated noise components at the receivers’ observations and prove its tightness under a more capable broadcast channel assumption. Our results demonstrate how optimized binning schemes simultaneously achieve high perceptual quality, low distortion, and strong secrecy, illuminating fundamental information-theoretic limits for next-generation trustworthy computation systems.

I. INTRODUCTION

Modern communication networks increasingly emphasize the transmission of *meaningful* information rather than aiming for exact bit-level signal fidelity, thus driving significant interest in *semantic communications*. Instead of reconstructing signals perfectly, semantic systems focus on preserving only the salient features of the data that directly influence end applications [1], [2]. This paradigm is especially crucial in bandwidth- and latency-constrained environments, such as immersive multimedia and autonomous driving, where transmitting all raw data can be prohibitively expensive.

From an information-theoretic perspective, semantic communications can be viewed as a *remote source coding* problem [3]–[5], where the receiver aims to approximate or compute a specific function of the source. In our recent work, *randomized distributed function computation* (RDFC) framework [6] is introduced to incorporate a controlled randomization step at the encoder. This randomization produces synthesized outputs with a desired statistical distribution while preserving essential functionalities. Applications of RDFC include neural compression via generative models [7], [8], federated learning with side information [9], and differential privacy mechanisms [6], [10], [11]. Based on the strong coordination results [12], the RDFC framework maintains strong performance guarantees, also for minimal or no common randomness [6], [13], [14].

We focus on the *rate-distortion-perception* (RDP) problem, which is highly relevant for image compression [15, Section 17.4.2], [16]–[21]. The RDP problem aims to both minimize the expected distortion between the original and reconstructed images and ensure that the reconstructed images follow a probability distribution close to that of the original input, the latter of which preserves perceptual realism. Such

realism is enforced by ensuring the reconstructed distribution approximates the source distribution [22], [23], a notion supported by deep learning-based generative discriminators [24]–[26]. We study two core settings: a *noiseless* transmission model, which can abstract image compression at higher open systems interconnection layers, and a *noisy* broadcast channel model, which can correspond to a multi-layer scenario. In the noiseless setting, the entire transmission can be viewed as end-to-end image compression, directly fitting into the RDFC structure by mapping encoder outputs to decoder inputs without channel impairments. In contrast, when the source is sent over a broadcast channel, noise correlation among multiple receivers’ channel observations becomes a general and realistic scenario. This added complexity demands a more involved analysis of the achievable rate, distortion, and perceptual realism trade-offs.

Furthermore, privacy and security are fundamental in practical deployments. The public transmission of encoder outputs may leak information to eavesdroppers, making physical layer security (PLS) techniques attractive for resilience against advanced threats [27], [28]. Methods such as joint source-channel coding (JSCC) for secure image compression [29] align with these RDP models, aiming to limit or even eliminate adversarial information gain. In this work, we derive both asymptotic characterizations of RDP under strong secrecy constraints. Specifically, we characterize the secure RDP region for noiseless transmission channels by establishing the optimal RDP trade-offs under a strong secrecy constraint. We further derive an inner bound on the secure RDP region for noisy broadcast channel transmissions, accounting for correlated channel noise components and applying a refined random binning approach to provide strong secrecy guarantees. Moreover, under a more capable channel constraint, we provide a tight characterization of the secure RDP region over noisy channels.

Notation: Denote total variation distance between two probability distributions as $\|P_Y - P_X\|_{\text{TV}} \triangleq 0.5 \sum_{b \in \mathcal{B}} |P_Y(b) - P_X(b)|$. Denote the set of integers $\{a, a+1, \dots, b\}$ as $[a : b]$. $X \sim \text{Unif}[a : b]$ represents a random variable X that is uniformly distributed on the interval between a and b .

II. SYSTEM MODELS

Consider an encoder observing an input sequence $X^n \in \mathcal{X}^n$ i.i.d. according to Q_X , and common randomness $C \in [1 : 2^{nR_0}]$ shared with a decoder. The encoder outputs an index $S \in [1 : 2^{nR}]$ as a function of its inputs. The common randomness C is independent of X^n and uniformly distributed,

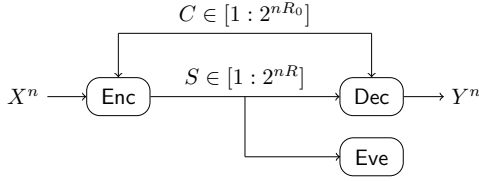


Fig. 1. System model for the secure RDP problem for transmissions over noiseless channels.

i.e., $C \sim P_C = \text{Unif}[1 : 2^{nR_0}]$. The decoder observes the index S and the common randomness C . It outputs a sequence $Y^n \in \mathcal{X}^n$ such that (i) the induced output distribution of Y^n is close in total variation to Q_X^n , and (ii) the expected distortion between the input X^n and output Y^n sequences is minimized. Moreover, an eavesdropper (Eve) observes the public index S . Thus, the encoder-decoder pair aims to ensure that S is independent of Y^n . This secrecy leakage scenario can refer to applications like generative artificial intelligence used to produce artistic digital content, where the reconstructed output is the end product. Fig. 1 depicts the model for this problem, which corresponds to the scenario with transmission over noiseless channels.

Next, we define the secure RDP region for the transmission of the index over noiseless channels.

Definition 1. A secure RDP tuple (R, R_0, D) is achievable for an i.i.d target distribution Q_X^n if, for any $\epsilon > 0$, there exist $n \geq 1$, an encoder, and a decoder such that

$$\|P_{Y^n} - Q_X^n\|_{TV} \leq \epsilon \quad (\text{realism}) \quad (1)$$

$$I(Y^n; S) \leq \epsilon \quad (\text{secrecy}) \quad (2)$$

$$\mathbb{E}[d(X^n, Y^n)] \leq D + \epsilon \quad (\text{distortion}) \quad (3)$$

with a per-letter distortion metric $d(x^n, y^n) = \frac{1}{n} \sum_{i=1}^n d(x_i, y_i)$. The secure RDP region $\mathcal{R}$ is the closure of the set of all achievable tuples. $\diamond$

We extend the first model by considering transmission over noisy broadcast channels, where the noise components on the channel observations of different receivers are correlated. The second model is depicted in Fig. 2. In this model, the encoder observes an input $X^n \in \mathcal{X}^n$, which is i.i.d. according to Q_X , and common randomness $C \in [1 : 2^{nR_0}]$ to produce a channel input $\tilde{X}^n \in \tilde{\mathcal{X}}^n$ that is sent over a memoryless broadcast channel $P_{\tilde{Y}\tilde{Z}|\tilde{X}}$. The common randomness C is independent of X^n and uniformly distributed. The decoder observes the channel output $\tilde{Y}^n$ and the common randomness C to output a sequence $Y^n \in \mathcal{X}^n$ such that (i) the induced output distribution of Y^n is close in total variation to Q_X^n , and (ii) the distortion between the input and output sequences is minimized. Similar to the first model, an Eve observes another channel output sequence $\tilde{Z}^n$, which should be independent of Y^n .

Next, we define the secure noisy RDP region for the transmission of the index over noisy broadcast channels.

Definition 2. A secure noisy RDP tuple (R, R_0, D) is achievable for an i.i.d target distribution Q_X^n and memoryless broad-

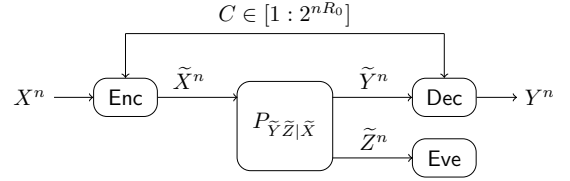


Fig. 2. System model for the secure noisy RDP problem for transmissions over noisy broadcast channels.

cast channel $P_{\tilde{Y}\tilde{Z}|\tilde{X}}$ if, for any $\epsilon > 0$, there exist $n \geq 1$, an encoder, and a decoder such that (1), (3), and

$$I(Y^n; \tilde{Z}^n) \leq \epsilon \quad (\text{secrecy}) \quad (4)$$

with a per-letter distortion metric $d(\cdot, \cdot)$. The secure noisy RDP region $\mathcal{R}_n$ is the closure of the set of all achievable tuples. $\diamond$

We remark that the secrecy constraints in (2) and (4) provide strong secrecy, i.e., the unnormalized leakage to Eve is negligible.

Next, we define more capable broadcast channels, assumed to establish a tight secure noisy RDP region below.

Definition 3. $P_{\tilde{Y}\tilde{Z}|\tilde{X}}$ is more capable if, for all $P_{\tilde{X}}$, we have

$$I(\tilde{Y}; \tilde{X}) \geq I(\tilde{Z}; \tilde{X}). \quad (5)$$

III. SECURE RDP RATE REGIONS OVER CHANNELS

We characterize the secure RDP region $\mathcal{R}$, for the system model depicted in Fig. 1, in Theorem 1. The proof of Theorem 1 is given in Section IV.

Theorem 1. The region $\mathcal{R}$ is the union overall joint probability distributions P_{XUY} of the set of all (R, R_0, D) tuples that satisfy

$$R \geq I(U; X), \quad (6)$$

$$R_0 \geq I(U; Y), \quad (7)$$

$$D \geq \mathbb{E}[d(X, Y)] \quad (8)$$

such that $X-U-Y$ forms a Markov chain, and the cardinality $|\mathcal{U}|$ of U can be limited to $|\mathcal{U}| \leq |\mathcal{X}|^2 + 1$.

Next, we provide an inner bound on the secure noisy RDP region $\mathcal{R}_n$ in Theorem 2, whose proof is given in Section V.

Theorem 2. The rate region $\mathcal{R}_n$ for a noisy memoryless broadcast channel $P_{\tilde{Y}\tilde{Z}|\tilde{X}}$ includes the union overall joint probability distributions $P_{XW_1YW_2}$ of the set of all (R_0, D) tuples that satisfy (8) and

$$I(W_1; X) \leq I(W_2; \tilde{Y}), \quad (9)$$

$$R_0 \geq I(W_1; Y) + I(W_2; \tilde{Z}) - I(W_2; \tilde{Y}) \quad (10)$$

such that $X-W_1-Y$ and $W_2-\tilde{X}-(\tilde{Y}, \tilde{Z})$ form Markov chains, and (X, W_1, Y) are mutually independent of $(W_2, \tilde{X}, \tilde{Y}, \tilde{Z})$. It suffices to have $|\mathcal{W}_1| \leq |\mathcal{X}|^2 + 1$ and $|\mathcal{W}_2| \leq |\mathcal{X}| + 1$.

Next, we establish the secure noisy RDP region $\mathcal{R}_n$ when the decoder has a more capable channel than the eavesdropper's channel. Thus, we consider the scenarios, where we have (5) for all $P_{\tilde{X}}$. The proof of Corollary 1 is given in Section VI.

Corollary 1. *The region $\mathcal{R}_n$ for a decoder that has a more capable memoryless broadcast channel $P_{\tilde{Y}\tilde{Z}|\tilde{X}}$ than the eavesdropper is the union overall joint probability distributions P_{XWY} of the set of all (R_0, D) tuples that satisfy (8) and*

$$I(W; X) \leq I(\tilde{X}; \tilde{Y}), \quad (11)$$

$$R_0 \geq I(W; Y) + I(\tilde{X}; \tilde{Z}) - I(\tilde{X}; \tilde{Y}) \quad (12)$$

where $X - W - Y$ forms a Markov chain, (X, W, Y) and $(\tilde{X}, \tilde{Y}, \tilde{Z})$ are mutually independent, and $|\mathcal{W}| \leq |\mathcal{X}|^2 + 1$.

IV. PROOF OF THEOREM 1

Next, we provide the achievability and converse proofs for Theorem 1.

A. Achievability Proof of Theorem 1

Proof sketch. The achievability proof of Theorem 1 applies the output statistics of random binning (OSRB) [30] method.

Fix a distribution P_{XUY} such that $\mathbb{E}[d(X^n, Y^n)] \leq D + \epsilon_n$, in which $\epsilon_n > 0$ such that $\epsilon_n \rightarrow 0$ when $n \rightarrow \infty$. Generate an auxiliary random variable sequence U^n in an i.i.d manner such that we have a joint probability distribution $P_{XU} = Q_X P_{U|X}$. Suppose the encoder and decoder have access to another random variable F , drawn uniformly at random according to $P_F = \text{Unif}[1 : 2^{n\tilde{R}}]$. The random variable F represents the public choice of the encoder and decoder pair. Independently and uniformly assign bin indices C, F , and S to each sequence $u^n \in U^n$, which is done in two ways:

(i) *Random Binning* that (1) assigns $C = \varphi_1(U^n)$, where $\varphi_1 : U^n \rightarrow [1 : 2^{nR_0}]$ maps each sequence u^n uniformly at random and independently to an index $[1 : 2^{nR_0}]$; (2) assigns $F = \varphi_2(U^n)$, where $\varphi_2 : U^n \rightarrow [1 : 2^{n\tilde{R}}]$ maps each sequence u^n uniformly at random and independently to an index $[1 : 2^{n\tilde{R}}]$; and (3) assigns $S = \varphi_3(U^n)$, where $\varphi_3 : U^n \rightarrow [1 : 2^{nR}]$ maps each sequence u^n uniformly at random and independently to an index $[1 : 2^{nR}]$ such that the receiver can recover U^n from (C, F, S) using a decoder $P_{\hat{U}^n|CFS}^{\text{RB}}$ with the output $\hat{U}^n$; and

(ii) *Random Coding*, where the common randomness and the encoder-decoder choice are generated uniformly at random and independently according to P_C and P_F , respectively. The message index S is selected in the same way as for the random binning according to $P_{S|U^n}^{\text{RB}}$. U^n is recovered with the same decoder $P_{\hat{U}^n|CFS}^{\text{RB}}$, as in the random binning.

These two binning ways give rise to two different joint distributions, one for the random binning denoted as P^{RB} , and one for the random coding denoted as P^{RC}

$$\begin{aligned} P^{\text{RB}} &:= Q_X^n P_{U^n|X^n} P_{C|U^n} P_{F|U^n} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n} \\ &= P_{CFX^n} P_{U^n|CFX^n}^{\text{RB}} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n} \end{aligned} \quad (13)$$

$$P^{\text{RC}} := P_C P_F Q_X^n P_{U^n|CFX^n}^{\text{RB}} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n}. \quad (14)$$

The total variation distance between P^{RB} and P^{RC} can be upper-bounded via the following triangle inequality

$$\begin{aligned} & \|P^{\text{RB}} - P^{\text{RC}}\|_{\text{TV}} \\ & \leq \|P^{\text{RB}} - P_C P_F Q_X^n P_{U^n|CFX^n}^{\text{RB}} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n}\|_{\text{TV}} \\ & \quad + \|P_C P_F Q_X^n P_{U^n|CFX^n}^{\text{RB}} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n} - P^{\text{RC}}\|_{\text{TV}} \\ & \stackrel{(a)}{=} \|P^{\text{RB}} - P_C P_F Q_X^n P_{U^n|CFX^n}^{\text{RB}} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n}\|_{\text{TV}} \\ & \stackrel{(b)}{=} \|P_{CFX^n} - P_C P_F Q_X^n\|_{\text{TV}} \end{aligned} \quad (15)$$

where (a) follows from the definition of P^{RC} , (b) follows by [31, Lemma 17] since $P_{U^n|CFX^n}^{\text{RB}} P_{S|U^n}^{\text{RB}} P_{\hat{U}^n|CFS}^{\text{RB}} P_{Y^n|\hat{U}^n}$ appears in (13) as well. The total variation distance in (15) asymptotically tend to zero if we impose the rate constraints given below.

First, the public indices C and F are almost independent of X^n and jointly uniformly distributed if we have [30, Theorem 1]

$$\tilde{R} + R_0 < H(U|X). \quad (16)$$

The public message index S and the public index F are almost independent of Y^n and jointly uniformly distributed if we have [30, Theorem 1]

$$\tilde{R} + R < H(U|Y), \quad (17)$$

ensuring that the secrecy constraint in (2) is satisfied. Finally, we fix the $P_{U^n|CFX^n}^{\text{RB}}$, that's used in both P^{RC} and P^{RB} , to a Slepian-Wolf decoder $P_{\hat{U}^n|CFS}^{\text{SW}}$ that can reliably estimate U^n from (S, C, F) such that the sequence reconstruction error tends to zero as $n \rightarrow \infty$, if we have [30, Lemma 1]

$$\tilde{R} + R_0 + R > H(U). \quad (18)$$

Applying Fourier-Motzkin elimination [32, Appendix D], (16)-(18) results in the achievable rates

$$R = I(U; X) + \delta, \quad (19)$$

$$R_0 = I(U; Y) + \delta \quad (20)$$

for any $\delta > 0$ that tends to zero as $n \rightarrow \infty$.

Furthermore, the total variation distance between the induced output distribution of the codebook $U^n(c, s)$ and the output distribution in the realism constraint (1) is

$$\left\| \sum_{c,s} \frac{P_{Y^n|U^n}(y^n|U^n(c, s))}{2^{nR_0} 2^{nR}} - Q_X^n \right\|_{\text{TV}}. \quad (21)$$

The total variation distance between the assumed input distribution and the induced distribution of the codebook $U^n(c, s)$ is

$$\frac{1}{2^{nR_0}} \left\| \sum_{c,s} \frac{P_{Y^n|U^n}(y^n|U^n(c, s))}{2^{nR}} - Q_X^n \right\|_{\text{TV}}. \quad (22)$$

The rate of C is moved outside the total variation distance in (22) since it is independent of X^n . These total variation distances goes to zero if $R_0 + R > I(U; Y)$ and $R > I(U; X)$ according to the soft covering lemma [12, Lemma IV.1], which

is satisfied by (19) and (20), similar to the proof of Theorem 2 in [22].

With the P^{RB} and P^{RC} distributions being asymptotically equal by (15), [30, 3] in Lemma 4] guarantee the existence of an instance of $F = f$ such that when conditioning on this instance of $F = f$ the total variation distance between P^{RB} and P^{RC} in (15) is upper bounded as

$$\|P_{CFX^n}^{\text{RB}} - P_{CFX^n}^{\text{RC}}\|_{\text{TV}} \leq \|P_{CX^n|F=f}^{\text{RB}} - P_{CX^n|F=f}^{\text{RC}}\|_{\text{TV}} \quad (23)$$

which goes to zero when n goes to infinity by the imposed rates (19) and (20). The bounding of the total variation distance (29) and the reduction of the common randomness is done similar to [30, Part (3) in Section IV.B] and [33, Part (4) in Section IV.A]. However, our model also considers a realism constraint in (1) and a different secrecy constraint.

Moreover, since all sequence tuples (x^n, u^n, y^n) are in the jointly typical set with high probability, by the typical average lemma [32, pp. 26], the distortion constraint (3) is satisfied. $\square$

B. Converse Proof of Theorem 1

Proof sketch. Suppose there exists a tuple (R, R_0, D) for some $\epsilon > 0$ and $n \geq 1$ that satisfies (1)-(3). Define an auxiliary random variable $U_i \triangleq (S, C, X^{i-1})$ such that $X_i - U_i - Y_i$ forms a Markov chain and a time-sharing random variable $T \in [1 : n]$ that is independent of all other random variables.

We have

$$\begin{aligned} nR &\geq H(S|C) \geq I(X^n; S|C) \stackrel{(a)}{=} I(X^n; S, C) \\ &= \sum_{i=1}^n I(X_i; S, C|X^{i-1}) \stackrel{(b)}{=} \sum_{i=1}^n I(X_i; S, C, X^{i-1}) \\ &\stackrel{(c)}{=} nI(X_T; U_T|T) = nI(X_T; U_T, T) \stackrel{(d)}{=} nI(X; U) \end{aligned} \quad (24)$$

where (a) and (b) follow because X^n is independent of C and i.i.d, (c) follows from the definition of U_i and T , and (d) follows by defining $U \triangleq (U_T, T)$ and $X \triangleq X_T$.

We obtain

$$\begin{aligned} nR_0 &\geq H(C|S) \geq I(Y^n; C|S) \stackrel{(a)}{\geq} I(Y^n; C, S) - f(\epsilon) \\ &= \sum_{i=1}^n I(Y_i; C, S|Y^{i-1}) - f(\epsilon) \\ &= \sum_{i=1}^n [I(Y_i; C, S, Y^{i-1}) - I(Y_i; Y^{i-1})] - f(\epsilon) \\ &\stackrel{(b)}{\geq} \sum_{i=1}^n [I(Y_i; C, S, Y^{i-1}) - g(\epsilon)] - f(\epsilon) \\ &\stackrel{(c)}{\geq} \sum_{i=1}^n [I(Y_i; C, S, X^{i-1}) - g(\epsilon)] - f(\epsilon) \\ &\stackrel{(d)}{=} nI(Y_T; U_T|T) - 2ng(\epsilon) - f(\epsilon) \\ &= nI(Y_T; U_T, T) - 2ng(\epsilon) - f(\epsilon) \\ &\stackrel{(e)}{=} nI(Y; U) - 2ng(\epsilon) - f(\epsilon) \end{aligned} \quad (25)$$

where (a) follows for a function $f(\epsilon)$ that tends to zero as $\epsilon \rightarrow 0$, which follows from the secrecy constraint in (2), (b) follows by (1) as Y^n is nearly i.i.d and for a function $g(\epsilon)$ that tends to zero as $n \rightarrow \infty$, (c) follows by applying the data processing inequality to the Markov chain $(C, S, X^{i-1}) - (Y^{i-1}, C, S) - Y_i$, in (d) follows by the definition of U_i and T , and because Y^n is nearly i.i.d, and (e) follows by defining $U \triangleq (U_T, T)$ and $Y \triangleq Y_T$.

We have

$$\begin{aligned} D + \epsilon &\stackrel{(a)}{\geq} \mathbb{E}[d(X^n, Y^n)] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[d(X_i, Y_i)] \\ &\stackrel{(b)}{\geq} \mathbb{E}[d(X_T, Y_T)|T] + \epsilon' \stackrel{(c)}{=} \mathbb{E}[d(X, Y)] + \epsilon' \end{aligned} \quad (26)$$

where (a) follows by (3), (b) follows since Y^n is almost i.i.d for an ϵ' that goes to zero when $n \rightarrow \infty$ and (c) follows by defining $X \triangleq X_T$ and $Y \triangleq Y_T$.

Using the support lemma [4, Lemma 15.4], one can preserve the probability distribution P_{XY} by using $(|\mathcal{X}||\mathcal{Y}| - 1) = (|\mathcal{X}|^2 - 1)$ continuous real-valued functions. One should preserve two more expressions in Theorem 1, i.e., $I(U; X)$ and $I(U; Y)$. Thus, imposing $|\mathcal{U}| \leq |\mathcal{X}|^2 + 1$ suffices. $\square$

V. PROOF OF THEOREM 2

The achievability proof for the inner bound given in Theorem 2 follows by applying the OSRB method.

Proof sketch. Fix a distribution P_{XWY} such that $\mathbb{E}[d(X^n, Y^n)] \leq D + \epsilon_n$, where $\epsilon_n > 0$ such that $\epsilon_n \rightarrow 0$ when $n \rightarrow \infty$. Generate an auxiliary random variable sequence W^n in an i.i.d manner such that we have the joint probability distribution of $(X^n, W^n, \tilde{X}^n, \tilde{Y}^n, \tilde{Z}^n)$ as $P_{X^n W^n \tilde{X}^n \tilde{Y}^n \tilde{Z}^n} = Q_X^n P_{W|X}^n P_{\tilde{X}|WX}^n P_{\tilde{Y}Z|\tilde{X}}^n$ which, with a $W = (W_1, W_2)$, marginalizes as $P_{X^n W_1^n} = Q_X^n P_{W_1|X}^n$ and $P_{W_2^n \tilde{X}^n \tilde{Y}^n \tilde{Z}^n} = P_{W_2}^n P_{\tilde{X}|W_2}^n P_{\tilde{Y}Z|\tilde{X}}^n$. Suppose the encoder and the decoder have access to a random variable F , drawn uniformly at random according to a distribution $P_F = \text{Unif}[1 : 2^{n\tilde{R}}]$, which represents the choice of encoder and decoder. Independently and uniformly assign bin indices C and F at random to each sequence w^n , which is done in two ways:

(i) *Random binning* that 1) assigns $C = \varphi_1(W^n)$, where $\varphi_1 : \mathcal{W}^n \rightarrow [1 : 2^{nR_0}]$ maps each sequence w^n uniformly at random and independently to an index $[1 : 2^{nR_0}]$; and 2) assigns $F = \varphi_2(W^n)$, where $\varphi_2 : \mathcal{W}^n \rightarrow [1 : 2^{n\tilde{R}}]$ maps each sequence w^n uniformly at random and independently to an index $[1 : 2^{n\tilde{R}}]$ such that the receiver can recover W^n from $(C, F, \tilde{Y}^n)$ using a decoder $P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}}$ with the output $\tilde{W}^n$. The channel input $\tilde{X}^n$ is selected according to $P_{\tilde{X}^n|W^n X^n}^{\text{RB}}$; and

(ii) *Random coding*, where the common randomness and the encoder-decoder choice is selected uniformly at random and independently according to P_C and P_F , respectively. The channel input $\tilde{X}^n$ is selected according to the same distribution $P_{\tilde{X}^n|W^n X^n}^{\text{RB}}$ as in the random binning. Similarly, the decoder

recovers W^n by using the same decoder $P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}}$ as in the random binning.

Define P^{RB} as the distribution induced by the random binning and P^{RC} as the distribution induced by the random coding, we then have

$$\begin{aligned} P^{\text{RB}} &= Q_X^n P_{W^n|X^n} P_{C|W^n} P_{F|W^n} P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} \\ &\quad \cdot P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} P_{Y^n|\tilde{W}^n \tilde{Y}^n} \\ &= P_{CFX^n} P_{W^n|CFX^n} P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} \\ &\quad \cdot P_{Y^n|\tilde{W}^n \tilde{Y}^n}, \end{aligned} \quad (27)$$

$$\begin{aligned} P^{\text{RC}} &= P_C P_F Q_X^n P_{W^n|CFX^n} P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} \\ &\quad \cdot P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} P_{Y^n|\tilde{W}^n \tilde{Y}^n}. \end{aligned} \quad (28)$$

The total variation distance between P^{RB} and P^{RC} is upper-bounded by

$$\begin{aligned} \|P^{\text{RB}} - P^{\text{RC}}\|_{\text{TV}} &\leq \|P^{\text{RB}} - P_C P_F Q_X^n P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} \\ &\quad \cdot P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} P_{Y^n|\tilde{W}^n \tilde{Y}^n}\|_{\text{TV}} \\ &\quad + \|P_C P_F Q_X^n P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} \\ &\quad \cdot P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} P_{Y^n|\tilde{W}^n \tilde{Y}^n} - P^{\text{RC}}\|_{\text{TV}} \\ &\stackrel{(a)}{=} \|P^{\text{RB}} - P_C P_F Q_X^n P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} \\ &\quad \cdot P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} P_{Y^n|\tilde{W}^n \tilde{Y}^n}\|_{\text{TV}} \\ &\stackrel{(b)}{=} \|P_{CFX^n}^{\text{RB}} - P_C P_F Q_X^n\|_{\text{TV}} \end{aligned} \quad (29)$$

where (a) follows by the definition of P^{RC} in (28), (b) follow by [31, Lemma 17] since $P_{\tilde{X}^n|W^n X^n}^{\text{RB}} P_{\tilde{Y}^n \tilde{Z}^n|\tilde{X}^n} P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}} P_{Y^n|\tilde{W}^n \tilde{Y}^n}$ appears in (27) as well.

The total variation distance in (29) will go to zero if we impose the following rates. The public index F and the common randomness index C are jointly uniformly distributed and independent of X^n if we have [30, Theorem 1]

$$\tilde{R} + R_0 < H(W|X). \quad (30)$$

Next, we fix the decoder $P_{\tilde{W}^n|CF\tilde{Y}^n}^{\text{RB}}$ to be a Slepian-Wolf decoder which can reliably recover W^n from $(C, F, \tilde{Y}^n)$ such that the sequence reconstruction error tends to zero as $n \rightarrow \infty$, if we have [30, Lemma 1]

$$\tilde{R} + R_0 > H(W|\tilde{Y}). \quad (31)$$

Finally, F is independent of $(Y^n, \tilde{Z}^n)$ if we have [30, Theorem 1]

$$\tilde{R} < H(W|Y, \tilde{Z}) \quad (32)$$

which satisfies the secrecy constraint in (4).

Suppose $W = (W_1, W_2)$ such that (X, W_1, Y) are mutually independent of $(W_2, \tilde{X}, Y, \tilde{Z})$. By (31) and (32), we have

$$\begin{aligned} R_0 &> H(W|\tilde{Y}) - H(W|Y, \tilde{Z}) \\ &= I(W_1, W_2; Y, \tilde{Z}) - I(W_1, W_2; \tilde{Y}) \\ &= I(W_1; Y) + I(W_2; \tilde{Z}) - I(W_2; \tilde{Y}). \end{aligned} \quad (33)$$

Moreover, by (30) and (31), we have

$$I(W; \tilde{Y}) = I(W_2; \tilde{Y}) > I(W_1; X) = I(W; X). \quad (34)$$

Similar to the arguments in [34, Section IV], the independence between RDP and channel variables and the strong coordination of $(Y^n, \tilde{Z}^n)$ ensure that the secrecy constraint in (4) is satisfied.

Let R_j represent the reliable communication rate between the encoder and decoder without any secrecy constraint. The total variation distance between the induced distribution from the codebook $W_1(c, j)$ and the target output distribution of the realism constraint (1) is

$$\left\| \sum_{c,j} \frac{P_{Y^n|W_1^n}(y^n|W_1(c, j))}{2^{nR_j} 2^{nR_0}} - Q_X^n \right\|_{\text{TV}}. \quad (35)$$

The total variation distance between the assumed input distribution and the induced distribution of the codebook $W_1(c, s)$ is

$$\frac{1}{2^{nR_0}} \left\| \sum_{c,j} \frac{P_{X^n|W_1^n}(x^n|W_1(c, j))}{2^{nR_j}} - Q_X^n \right\|_{\text{TV}}. \quad (36)$$

The total variation distances above goes to zero when $n \rightarrow \infty$ if we impose the following rates from [12, Lemma IV.1]

$$R_J + R_0 > I(W_1; Y), \quad (37)$$

$$R_J > I(W_1; X). \quad (38)$$

Define the rate of reliable communication between the encoder and decoder as $R_J = I(W_2; \tilde{Y})$, which is achievable as we have

$$nI(W_2; \tilde{Y}) = I(W_2^n; \tilde{Y}^n) \stackrel{(a)}{\leq} I(\tilde{X}^n; \tilde{Y}^n) = nI(\tilde{X}; \tilde{Y}) \quad (39)$$

where (a) follows by the data processing inequality applied to the Markov chain $W_2^n - \tilde{X}^n - \tilde{Y}^n$. By (34) and from the definition of R_J , (38) is satisfied. Moreover, from the definition of R_J , (37) is satisfied by imposing (33). Furthermore, since all sequence tuples (x^n, w^n, y^n) are in the jointly typical set with high probability, by the typical average lemma [32, pp. 26], the distortion constraint (3) is satisfied.

In the same way as the proof for Theorem 1, the total variation distance in (29) goes to zero ensures the existence of an instance $F = f$ such that the two distributions P^{RB} and P^{RC} goes to zero as n tends to infinity while conditioning on this instance of $F = f$. Moreover, as in Section IV, the cardinality bounds follow straightforwardly using the support lemma, given the Markov chains imposed. Note that our proof steps are similar to [33], but their model does not consider realism or distortion constraints, resulting in different regions. Our binning structure above is also different from the structure in [33] and simplifies the proof steps.

VI. PROOF OF COROLLARY 1

The achievability proof for Corollary 1 follows from the achievability proof for Theorem 2 with $W_1 = W$ and $W_2 = \tilde{X}$.

For the converse, suppose there exists a tuple (R_0, D) for some $\epsilon > 0$ and $n \geq 0$ that satisfies (1), and (3)-(5). Define an auxiliary random variable $W_t = (C, \tilde{Y}^n)$ and a time-sharing random variable $T \in [1 : n]$ that is independent of $(X^n, Y^n, C, \tilde{X}^n, \tilde{Y}^n, \tilde{Z}^n)$. We have

$$\begin{aligned}
0 &\stackrel{(a)}{\leq} I(\tilde{X}^n; \tilde{Y}^n) - I(\tilde{Y}^n; X^n, C) \\
&\stackrel{(b)}{=} nI(\tilde{X}; \tilde{Y}) - \sum_{t=1}^n I(\tilde{Y}^n; X_t | X^{t-1}, C) \\
&\stackrel{(c)}{=} nI(\tilde{X}; \tilde{Y}) - \sum_{t=1}^n I(\tilde{Y}^n, X^{t-1}, C; X_t) \\
&\leq nI(\tilde{X}; \tilde{Y}) - \sum_{t=1}^n I(\tilde{Y}^n, C; X_t) \\
&\stackrel{(d)}{=} n[I(\tilde{X}; \tilde{Y}) - I(W_T; X_T | T)] \\
&\stackrel{(e)}{=} n[I(\tilde{X}; \tilde{Y}) - I(W; X)] \quad (40)
\end{aligned}$$

where (a) follows from the data-processing inequality applied to the Markov chain $(X^n, C) - \tilde{X}^n - \tilde{Y}^n$, (b) follows because the channel $P_{\tilde{Y}|\tilde{X}}$ is memoryless, (c) follows because X^n is i.i.d and independent of C , (d) follows from the definition of W_t and T , and (e) follows from defining $W \triangleq (W_T, T)$ and $X \triangleq X_T$.

We observe

$$\begin{aligned}
nR_0 &\geq H(C|\tilde{Y}^n) - H(C|\tilde{Y}^n, Y^n) = I(Y^n; C|\tilde{Y}^n) \\
&\stackrel{(a)}{\geq} I(Y^n; C, \tilde{Y}^n) - I(Y^n; \tilde{Y}^n) + I(Y^n; \tilde{Z}^n) - f(\epsilon) \quad (41)
\end{aligned}$$

where (a) follows by (4) where $f(\epsilon)$ is a function that tends to zero as $\epsilon \rightarrow 0$.

For the first term in (41), we have

$$\begin{aligned}
I(Y^n; C, \tilde{Y}^n) &= \sum_{t=1}^n I(Y_t; C, \tilde{Y}^n | Y^{t-1}) \\
&\stackrel{(a)}{\geq} \sum_{t=1}^n I(Y_t; C, \tilde{Y}^n, Y^{t-1}) - ng(\epsilon) \\
&\geq \sum_{t=1}^n I(Y_t; C, \tilde{Y}^n) - ng(\epsilon) \stackrel{(b)}{=} nI(Y_T; W_T | T) - 2ng(\epsilon) \\
&= n[I(Y_T; W_T, T) - I(Y_T; T)] - 2ng(\epsilon) \\
&\stackrel{(c)}{\geq} nI(Y_T; W_T, T) - 3ng(\epsilon) \stackrel{(d)}{=} nI(Y; W) - 3ng(\epsilon) \quad (42)
\end{aligned}$$

where (a) follows from [34, Lemma 3] since Y^n is nearly i.i.d by (1), where $g(\epsilon)$ is a function that tends to zero as $\epsilon \rightarrow 0$, (b) is from the definition of W_t and T , and because Y^n is nearly i.i.d, (c) follow from [12, Lemma VI.3] and by (1), and (d) follows by defining $Y \triangleq Y_T$.

For the second and third terms in (41), we have

$$\begin{aligned}
&I(Y^n; \tilde{Y}^n) - I(Y^n; \tilde{Z}^n) \\
&\stackrel{(a)}{=} \sum_{t=1}^n [I(Y^n; \tilde{Y}_t | \tilde{Y}^{t-1}, \tilde{Z}_{t+1}^n) - I(Y^n; \tilde{Z}_t | \tilde{Y}^{t-1}, \tilde{Z}_{t+1}^n)] \\
&\stackrel{(b)}{=} \sum_{t=1}^n [I(W_{2,t}; \tilde{Y}_t | W_{3,t}) - I(W_{2,t}; \tilde{Z}_t | W_{3,t})] \\
&\stackrel{(c)}{\leq} n[I(W_{2,T}; \tilde{Y}_T | W_{3,T}, T) - I(W_{2,T}; \tilde{Z}_T | W_{3,T}, T)] + ng'(\epsilon) \\
&\stackrel{(d)}{=} n[I(W_2; \tilde{Y}_T | W_3) - I(W_2; \tilde{Z}_T | W_3)] + ng'(\epsilon) \\
&\leq n \max_t [I(W_2; \tilde{Y}_T | W_3 = t) - I(W_2; \tilde{Z}_T | W_3 = t)] + ng'(\epsilon) \\
&\stackrel{(e)}{=} n[I(\tilde{W}_2; \tilde{Y}) - I(\tilde{W}_2; \tilde{Z})] + ng'(\epsilon) \\
&\stackrel{(f)}{\leq} n[I(\tilde{X}; \tilde{Y}) - I(\tilde{X}; \tilde{Z})] + ng'(\epsilon) \quad (43)
\end{aligned}$$

where (a) follows from Csiszár's sum identity [32, pp. 25], (b) follows by defining $W_{3,t} \triangleq (\tilde{Y}^{t-1}, \tilde{Z}_{t+1}^n)$ and $W_{2,t} \triangleq (Y^n, W_{3,t})$, (c) follows for a function $g'(\epsilon)$ such that $g'(\epsilon) \rightarrow 0$ when $\epsilon \rightarrow 0$ since Y^n is almost i.i.d by (1), and defining $W_2 \triangleq (W_{2,T}, T)$ and $W_3 \triangleq (W_{3,T}, T)$, (d) follows by defining $\tilde{Y} \triangleq \tilde{Y}_T$, $\tilde{Z} \triangleq \tilde{Z}_T$, (e) follows by $\tilde{W}_2 \sim P_{W_2|W_3=t^*}$ (where t^* is the maximizer) and that $\tilde{W}_2 - \tilde{X} - (\tilde{Y}, \tilde{Z})$ forms a Markov chain, and (f) follows from the more capable channel assumption. Combining (41)-(43) results in (12).

The distortion bound follows as in (26). Moreover, the cardinality bound follows by using the support lemma, as in Section IV. We remark that this converse proof follows similarly to [34], with the difference that our model also considers realism and distortion constraints. Moreover, the secrecy constraints imposed are different. $\square$

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